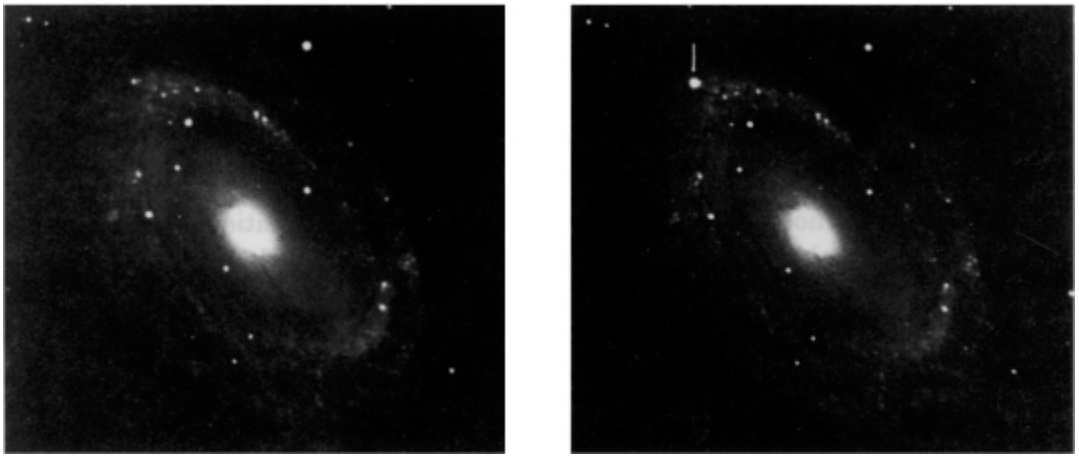
nebulas were not simply clouds of gas in our own Milky Way galaxy, as had been thought, but rather were galaxies in their own right-giant assemblages of nearly 10^{12} (a trillion) stars, far outside our own galaxy. The rare novae seen in these galaxies, being so much farther away than our own galaxy's ordinary novae, would have to be intrinsically far more luminous than ordinary novae in order to have a similar brightness as seen from Earth.

Baade collected from the published literature all the observational data he could find about each of the six superluminous novae that astronomers had seen since the turn of the century. These data he combined with all the observational information he could get about the distances to the galaxies in which they lay, and from this combination he computed how much light the superluminous novae put out. His conclusion was startling: During flare-up these superluminous novae were typically 10^8 (100 million) times more luminous than our Sun! (Today we know, thanks largely to work in 1952 by Baade himself, that the distances were underestimated in the 1930s by nearly a factor of 10, and that correspondingly[1] the superluminous novae are nearly 10^{10}—10 billion—times more luminous than our Sun.)



The galaxy NGC 4725 in the constellation Coma Berenices. .*Left:* As photographed on 10 May 1940, before a supernova outburst. *Right:* On 2 January 1941 during the supernova outburst. The white line points to the supernova, in the outer reaches of the galaxy. This galaxy is now known to be 30 million light-years from Earth and to contain 3×10^{11} (a third of a trillion) stars. [Courtesy California Institute of Technology.]

Zwicky, a lover of extremes, was fascinated by these superluminous novae. He and Baade discussed them endlessly and coined for them the name *supernovae.* Each supernova, they presumed (correctly), was